

Software Requirements Specification (SRS)

PhishGuard: An Explainable Browser Extension for Real-Time Phishing Detection

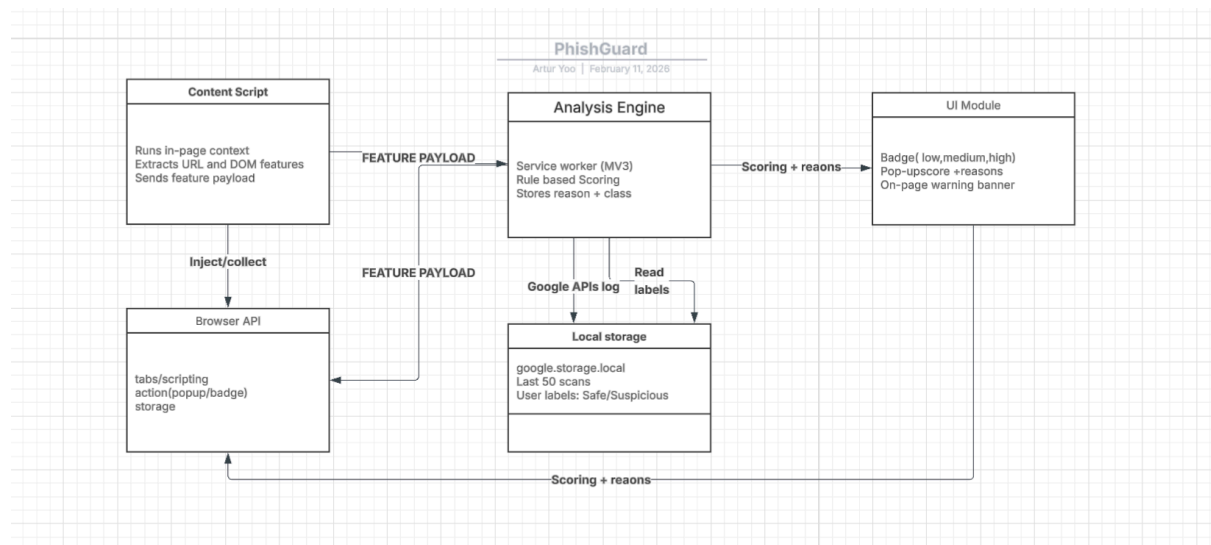
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5.1 Requirements Introduction

PhishGuard is a Google Chrome browser extension that analyzes webpages in real time to detect phishing indicators and present a risk classification to the user before credentials are submitted. The system extracts URL features and page-level structural features, computes a deterministic phishing risk score, and communicates risk classification through browser-based user interface components.



5.2 Functional Requirements

Functional requirements describe what the system shall do, independent of implementation details. Each requirement is written as a discrete, testable statement using “shall” form, as required by the CMSI 4072 guidelines .

5.2.1 URL Feature Extraction

Description:

The system shall analyze the structure and composition of the active webpage URL.

Requirements:

- 5.2.1.1 The system shall extract the full URL of the active tab.
 - 5.2.1.2 The system shall parse and store the hostname of the URL.
 - 5.2.1.3 The system shall determine whether the hostname is an IPv4 address.
 - 5.2.1.4 The system shall compute the number of subdomains in the hostname.
 - 5.2.1.5 The system shall detect the presence of punycode encoding in the hostname.
 - 5.2.1.6 The system shall compare the URL string against a predefined list of at least 20 phishing-related keywords.
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5.2.2 Page Structure Analysis

Description:

The system shall analyze DOM-level features associated with phishing behavior.

Requirements:

- 5.2.2.1 The system shall detect the presence of at least one `<input type="password">` field.
 - 5.2.2.2 The system shall detect all HTML `<form>` elements on the page.
 - 5.2.2.3 The system shall determine whether any form submission target domain differs from the page's hostname.
 - 5.2.2.4 The system shall extract the page title string for keyword comparison.
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5.2.3 Risk Scoring Engine

Description:

The system shall compute a phishing risk score using deterministic rule-based weighting.

Requirements:

5.2.3.1 The system shall assign a numeric weight to each detected phishing indicator.

5.2.3.2 The system shall compute a total risk score between 0 and 100.

5.2.3.3 The system shall classify risk as:

- Low: 0–29
- Medium: 30–59
- High: 60–100

5.2.3.4 The system shall store the contributing indicators for each computed score.

5.2.4 Warning and Notification Interface

Description:

The system shall communicate risk results clearly to the user.

Requirements:

5.2.4.1 The system shall display an on-page warning banner when risk classification is High.

5.2.4.2 The system shall display a colored extension badge indicating current risk classification.

5.2.4.3 The system shall provide a popup interface displaying:

- Numeric risk score
- Risk classification
- List of contributing indicators

5.2.4.4 The system shall allow the user to dismiss warning banners.

5.2.5 Scan Logging and Feedback

Description:

The system shall support local logging of scan results.

Requirements:

5.2.5.1 The system shall store the 50 most recent scan results in local browser storage.

5.2.5.2 The system shall allow the user to label a scanned page as “Safe” or “Suspicious.”

5.2.5.3 The system shall display stored scan results in the popup interface.

5.2.6 Privacy Constraints

Description:

The system shall operate without collecting sensitive user data.

Requirements:

- 5.2.6.1 The system shall perform all feature analysis locally within the browser.
 - 5.2.6.2 The system shall not capture or store user-entered credentials.
 - 5.2.6.3 The system shall not transmit browsing data to external servers.
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5.3 Performance Requirements

Performance requirements define operational constraints on system behavior.

5.3.1 Page Analysis Latency

- 5.3.1.1 The system shall complete feature extraction and risk computation within 2000 milliseconds of page load completion.
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5.3.2 Resource Utilization

- 5.3.2.1 The system shall not consume more than 100MB of RAM during operation.
 - 5.3.2.2 The system shall not increase page load time by more than 5% under standard browsing conditions.
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5.3.3 Storage Limits

- 5.3.3.1 The system shall limit stored scan records to a maximum of 50 entries.
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5.4 Environment Requirements

5.4.1 Development Environment Requirements

- Operating System: macOS, Windows, or Linux
- Browser: Google Chrome (latest stable version)
- Programming Language: JavaScript or TypeScript
- IDE: VS Code or equivalent

- Version Control: Git
 - Testing Scripts: Python 3.x for dataset evaluation
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5.4.2 Execution Environment Requirements

- Chrome or Chromium-based browser supporting Manifest V3
- Desktop or laptop computer
- Minimum 4GB RAM
- Internet connection for browsing